

Supplementary information

Clustering trajectories of modelled estimates: an uncertainty aware framework, with an application to global cardiovascular disease burden, 1990 to 2023

Eben Afrifa-Yamoah

School of Science, Edith Cowan University, Perth, Western Australia

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This document contains the country-level results, the sensitivity tables referred to in the main text, a worked description of the data extraction, and the reproduction instructions. Section and table numbers are prefixed S. Colour conventions follow the main text: orange is the stalled cluster, blue the sustained-decline cluster, green the third cluster found only in the narrow-interval analysis, and dark grey the unassigned class.

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S1 Data extraction

S1.1 Connector calls

All series were retrieved through the IHME GBD 2023 Model Context Protocol connector with the `get_multi_location_trend` tool. The connector caps a single response by token budget rather than by row count, so the number of rows returned depends on the output format and the measure: about 667 location-years for DALYs in JSON format, about 476 for deaths in JSON format (the deaths output carries more columns), and about 714 for deaths in CSV format. Each global series was therefore assembled from twelve calls, partitioned by World Bank income group and year window as in Table S1. The four income groups partition the 203 countries that carry a World Bank code; Hong Kong and Macao appear under their own codes but sit under China in the GBD hierarchy and were dropped, and Tuvalu carries no code in the connector.

Table S1: Extraction schedule. Each cell is one connector call; windows were sized so that every call stays below the response cap.

World Bank income group	Countries	Year windows
Low income	26	1990 to 2006; 2007 to 2023
Lower middle income	51	1990 to 2001; 2002 to 2013; 2014 to 2023
Upper middle income	54	1990 to 2001; 2002 to 2013; 2014 to 2023
High income	72	1990 to 1998; 1999 to 2007; 2008 to 2016; 2017 to 2023

Two fields were retained per country-year: the age-standardised rate for both sexes, and the all-ages rate with its 95% uncertainty interval. The connector does not return an interval for the age-standardised rate, and does not expose the posterior draws from which the published intervals are computed. The Socio-demographic Index for 2023 was retrieved for all 204 locations in one call, and the GBD super region of each country was taken from the location hierarchy returned by the ranking tool.

S1.2 Interval width as a measurement quality proxy

The quantity w_i of main-text equation (2) is the country mean of the per-observation log-scale standard deviation implied by the published interval of the all-ages DALY rate. Figure 7a of the main text shows its distribution by super region. The median is 0.024 in Central and Eastern Europe and Central Asia, 0.033 in Latin America and the Caribbean, 0.038 in the high-income super region, 0.067 in North Africa and the Middle East, 0.069 in Southeast Asia, East Asia and Oceania, 0.097 in South Asia and 0.107 in sub-Saharan Africa. The proxy reflects every source of modelling uncertainty, not only the sparseness of primary data, and would be replaced by GBD’s published vital registration star ratings in a final analysis.

S2 Country-level results

Table S2 lists every country in the analysis set with its super region, World Bank income group, 2023 SDI, percentage change in the age-standardised DALY and death rates from 1990 to 2023, year of the cardiovascular DALY peak, interval width, and the assigned cluster and membership probability under the primary analysis, the deaths analysis and the 2000 to 2023 window. Cluster 1 is stalled, cluster 2 sustained decline, and 0 unassigned (membership probability below 0.70).

Table S2: Country level results. Region codes: CEECA, Central Europe, Eastern Europe and Central Asia; HI, high income; LAC, Latin America and Caribbean; NAME, North Africa and Middle East; SA, South Asia; SEAO, Southeast Asia, East Asia and Oceania; SSA, sub-Saharan Africa. Inc.: World Bank income group (L low, LM lower middle, UM upper middle, H high). Δ DALY and Δ deaths: percentage change in the age-standardised rate, 1990 to 2023. Peak: year of maximum centred DALY trajectory. w_i : mean log-scale standard deviation of the published interval. a_i : assigned cluster (1 stalled, 2 sustained decline, 0 unassigned) and p_i : membership probability, for the primary DALY analysis; superscript d, the deaths analysis; superscript w, the 2000 to 2023 window.

Country	Region	Inc.	SDI	Δ DALY	Δ deaths	Peak	w_i	a_i	p_i	a_i^d	p_i^d	a_i^w	p_i^w
Albania	CEECA	UM	0.7068	-41.5	-37.8	1990	0.043	0	0.65	0	0.34	2	0.85
Armenia	CEECA	UM	0.7173	-51.6	-50.5	1993	0.027	2	0.99	2	0.92	2	1.00
Azerbaijan	CEECA	UM	0.7053	-32.5	-29.1	1994	0.041	0	0.60	1	0.89	2	0.83
Belarus	CEECA	UM	0.8036	-23.1	-21.6	1998	0.018	0	0.69	0	0.29	2	0.99
Bosnia and Herzegovina	CEECA	UM	0.7129	-40.7	-37.9	1990	0.033	0	0.64	0	0.33	2	0.79
Bulgaria	CEECA	H	0.7708	-34.8	-34.3	1997	0.018	0	0.66	0	0.35	2	0.84
Croatia	CEECA	H	0.7821	-56.2	-54.6	1990	0.021	2	1.00	2	0.95	2	1.00
Czechia	CEECA	H	0.8310	-64.7	-63.1	1990	0.024	2	1.00	2	1.00	2	1.00
Estonia	CEECA	H	0.8549	-64.3	-62.4	1994	0.023	2	1.00	2	1.00	2	1.00
Georgia	CEECA	UM	0.7546	-48.4	-48.9	1991	0.026	2	0.94	2	0.79	2	1.00
Hungary	CEECA	H	0.7935	-49.8	-45.7	1991	0.019	2	1.00	2	0.91	2	0.88
Kazakhstan	CEECA	UM	0.7272	-42.8	-44.8	2006	0.021	2	0.92	2	0.90	2	1.00

Table S2 continued

Country	Region	Inc.	SDI	Δ DALY	Δ deaths	Peak	w_i	a_i	p_i	a_i^d	p_i^d	a_i^w	p_i^w
Kyrgyzstan	CEECA	LM	0.6252	-32.9	-24.6	1995	0.023	2	0.82	0	0.17	2	1.00
Latvia	CEECA	H	0.8508	-42.5	-43.3	1994	0.020	2	0.96	2	0.88	2	1.00
Lithuania	CEECA	H	0.8562	-43.3	-42.0	1994	0.021	2	0.89	0	0.62	2	1.00
Mongolia	CEECA	UM	0.6597	-44.2	-43.9	1997	0.047	2	0.92	2	0.90	2	1.00
Montenegro	CEECA	UM	0.7886	-36.0	-35.1	1997	0.036	2	0.71	0	0.40	2	0.99
North Macedonia	CEECA	UM	0.7585	-42.2	-37.8	1995	0.026	2	0.82	0	0.43	2	1.00
Poland	CEECA	H	0.8470	-62.5	-60.8	1991	0.024	2	1.00	2	1.00	2	1.00
Republic of Moldova	CEECA	UM	0.7337	-41.3	-46.7	1995	0.020	2	0.91	2	0.91	2	1.00
Romania	CEECA	H	0.7687	-43.0	-42.5	1996	0.017	2	0.99	2	0.87	2	0.99
Russian Federation	CEECA	H	0.8225	-30.5	-36.4	2003	0.015	2	0.90	2	0.89	2	1.00
Serbia	CEECA	UM	0.7845	-46.7	-42.1	1996	0.030	2	0.99	2	0.87	2	1.00
Slovakia	CEECA	H	0.8131	-55.7	-52.7	1990	0.029	2	1.00	2	0.94	2	1.00
Slovenia	CEECA	H	0.8449	-67.7	-66.7	1990	0.028	2	1.00	2	1.00	2	1.00
Tajikistan	CEECA	LM	0.5512	-2.7	4.5	1994	0.052	1	0.93	1	1.00	1	0.96
Turkmenistan	CEECA	UM	0.6822	-14.0	-13.4	1996	0.024	0	0.43	1	0.83	2	0.80
Ukraine	CEECA	UM	0.7645	-30.1	-32.2	1995	0.019	2	0.71	0	0.32	2	1.00
Uzbekistan	CEECA	LM	0.6321	-18.0	-15.3	1995	0.021	1	0.88	1	0.93	0	0.43
Andorra	HI	H	0.8799	-43.6	-46.8	1990	0.080	2	0.89	2	0.92	2	0.76
Argentina	HI	UM	0.7647	-52.2	-53.3	1990	0.025	2	1.00	2	0.95	2	0.82
Australia	HI	H	0.8529	-67.2	-71.0	1990	0.039	2	1.00	2	1.00	2	1.00
Austria	HI	H	0.8540	-57.5	-57.6	1990	0.036	2	1.00	2	1.00	2	0.99
Belgium	HI	H	0.8770	-62.5	-63.7	1990	0.039	2	1.00	2	1.00	2	1.00
Brunei Darussalam	HI	H	0.8324	-45.7	-48.1	1991	0.054	2	0.96	2	0.94	2	0.87
Canada	HI	H	0.8873	-52.2	-57.1	1990	0.037	2	1.00	2	1.00	2	0.88
Chile	HI	H	0.7885	-55.5	-59.4	1990	0.026	2	1.00	2	0.98	2	0.98
Cyprus	HI	H	0.8558	-61.4	-59.7	1990	0.061	2	1.00	2	1.00	2	1.00
Denmark	HI	H	0.9160	-69.0	-69.8	1990	0.034	2	1.00	2	1.00	2	1.00
Finland	HI	H	0.8908	-59.2	-55.2	1990	0.034	2	1.00	2	0.98	2	1.00
France	HI	H	0.8611	-48.9	-56.2	1990	0.048	2	1.00	2	1.00	2	0.97
Germany	HI	H	0.8786	-55.9	-56.6	1990	0.039	2	1.00	2	1.00	2	0.95
Greece	HI	H	0.8312	-51.7	-55.4	1990	0.031	2	1.00	2	1.00	2	1.00
Greenland	HI	H	0.8602	-59.4	-57.4	1990	0.057	2	1.00	2	1.00	2	1.00
Iceland	HI	H	0.8890	-57.4	-55.6	1990	0.039	2	1.00	2	1.00	2	1.00
Ireland	HI	H	0.9069	-70.4	-68.9	1990	0.031	2	1.00	2	1.00	2	1.00
Israel	HI	H	0.8409	-70.2	-71.5	1992	0.038	2	1.00	2	1.00	2	1.00
Italy	HI	H	0.8344	-56.9	-57.8	1990	0.049	2	1.00	2	1.00	2	0.98
Japan	HI	H	0.8759	-50.7	-59.3	1990	0.048	2	1.00	2	1.00	2	0.84
Luxembourg	HI	H	0.8948	-68.5	-68.2	1990	0.033	2	1.00	2	1.00	2	1.00
Malta	HI	H	0.8305	-65.9	-65.9	1990	0.034	2	1.00	2	1.00	2	1.00
Monaco	HI	H	0.9227	-75.5	-76.5	1990	0.127	2	1.00	2	1.00	2	1.00
Netherlands	HI	H	0.9084	-63.4	-62.5	1990	0.036	2	1.00	2	1.00	2	1.00
New Zealand	HI	H	0.8664	-59.6	-58.2	1990	0.036	2	1.00	2	1.00	2	0.99
Norway	HI	H	0.9169	-68.1	-68.9	1990	0.042	2	1.00	2	1.00	2	1.00
Portugal	HI	H	0.7865	-68.4	-71.1	1990	0.035	2	1.00	2	1.00	2	1.00
Republic of Korea	HI	H	0.8917	-77.6	-78.6	1990	0.049	2	1.00	2	1.00	2	1.00
San Marino	HI	H	0.8830	-60.9	-65.6	1990	0.067	2	1.00	2	1.00	2	1.00
Singapore	HI	H	0.8700	-67.3	-69.6	1990	0.029	2	1.00	2	1.00	2	1.00
Spain	HI	H	0.8118	-57.2	-61.8	1990	0.041	2	1.00	2	1.00	2	0.99
Sweden	HI	H	0.9109	-59.7	-61.7	1990	0.039	2	1.00	2	1.00	2	1.00
Switzerland	HI	H	0.9461	-63.7	-62.1	1990	0.042	2	1.00	2	1.00	2	1.00
United Kingdom	HI	H	0.8804	-59.9	-62.2	1990	0.032	2	1.00	2	1.00	2	1.00
United States of America	HI	H	0.8765	-40.5	-45.8	1990	0.036	2	0.85	2	0.92	2	0.76
Uruguay	HI	H	0.7454	-50.7	-52.9	1990	0.027	2	1.00	2	0.97	2	0.94
Antigua and Barbuda	LAC	H	0.7559	-45.3	-47.0	1990	0.032	2	0.92	2	0.93	2	0.75
Bahamas	LAC	H	0.8136	-22.0	-18.2	1990	0.027	1	0.99	1	1.00	1	0.98
Barbados	LAC	H	0.7548	-37.3	-39.7	1990	0.031	2	0.76	2	0.82	1	0.78
Belize	LAC	UM	0.6285	-32.2	-32.1	2000	0.034	0	0.64	0	0.35	2	0.83
Bermuda	LAC	H	0.8269	-55.1	-56.9	1990	0.033	2	1.00	2	1.00	2	0.84
Bolivia	LAC	LM	0.6317	-36.1	-39.7	1991	0.112	0	0.34	0	0.33	0	0.64
Brazil	LAC	UM	0.6839	-52.0	-54.8	1990	0.024	2	1.00	2	0.97	2	0.98
Colombia	LAC	UM	0.6888	-52.9	-52.8	1990	0.024	2	1.00	2	0.95	2	0.84
Costa Rica	LAC	UM	0.7154	-36.6	-37.2	1993	0.027	2	0.88	2	0.91	2	0.78
Cuba	LAC	UM	0.6949	-36.2	-36.8	1990	0.024	0	0.38	0	0.35	1	0.90
Dominica	LAC	UM	0.7537	-22.5	-23.4	1992	0.046	1	0.99	1	0.96	1	1.00
Dominican Republic	LAC	UM	0.6581	-4.6	5.4	1990	0.042	1	1.00	1	1.00	1	1.00
Ecuador	LAC	UM	0.6910	-36.1	-34.1	1994	0.022	2	0.94	2	0.93	2	0.80
El Salvador	LAC	UM	0.6082	-28.3	-18.5	1990	0.046	0	0.18	1	1.00	1	1.00
Grenada	LAC	UM	0.6781	-42.2	-45.3	1991	0.034	2	0.83	2	0.93	0	0.56
Guatemala	LAC	UM	0.5538	-31.9	-21.3	1993	0.028	0	0.20	1	1.00	1	1.00
Guyana	LAC	H	0.6957	-43.2	-43.3	1990	0.030	2	0.72	0	0.41	0	0.47
Haiti	LAC	LM	0.4638	1.4	6.4	2023	0.100	1	1.00	1	1.00	1	1.00
Honduras	LAC	LM	0.5203	-10.4	4.2	1998	0.076	1	1.00	1	1.00	1	0.98

Table S2 continued

Country	Region	Inc.	SDI	Δ DALY	Δ deaths	Peak	w_i	a_i	p_i	a_i^d	p_i^d	a_i^w	p_i^w
Jamaica	LAC	UM	0.6919	-21.5	-26.2	1990	0.034	1	1.00	1	1.00	1	1.00
Mexico	LAC	UM	0.6934	-17.5	-17.0	1994	0.025	1	1.00	1	1.00	1	1.00
Nicaragua	LAC	LM	0.5357	-23.3	-13.8	1990	0.042	1	1.00	1	1.00	1	1.00
Panama	LAC	H	0.7154	-31.4	-32.2	1990	0.030	0	0.35	0	0.36	1	0.99
Paraguay	LAC	UM	0.6741	-21.5	-18.6	1990	0.039	1	0.99	1	1.00	1	1.00
Peru	LAC	UM	0.6856	-36.8	-38.8	1994	0.051	2	0.92	2	0.94	0	0.65
Puerto Rico	LAC	H	0.8475	-53.3	-55.9	1990	0.031	2	1.00	2	1.00	2	1.00
Saint Kitts and Nevis	LAC	H	0.7545	-52.8	-51.8	1990	0.027	2	1.00	2	0.96	2	0.86
Saint Lucia	LAC	UM	0.7029	-51.5	-53.6	1990	0.035	2	1.00	2	0.96	2	0.80
Saint Vincent and the Grenadines	LAC	UM	0.6585	-27.8	-31.3	1994	0.034	0	0.40	0	0.34	1	0.88
Suriname	LAC	UM	0.6549	-30.1	-30.9	1992	0.036	0	0.40	0	0.27	0	0.69
Trinidad and Tobago	LAC	H	0.7705	-42.9	-42.1	1990	0.023	2	0.94	2	0.93	2	0.79
United States Virgin Islands	LAC	H	0.8302	-52.6	-62.7	1990	0.034	2	1.00	2	1.00	2	0.94
Venezuela	LAC	UM	0.5802	-12.6	-9.7	1990	0.022	0	0.18	1	1.00	1	1.00
Afghanistan	NAME	L	0.3337	-29.0	-25.4	1990	0.070	0	0.27	1	0.88	1	0.72
Algeria	NAME	UM	0.6578	-39.2	-39.3	1990	0.081	0	0.57	0	0.35	0	0.44
Bahrain	NAME	H	0.7843	-65.7	-65.8	1990	0.050	2	1.00	2	1.00	2	1.00
Egypt	NAME	LM	0.6765	-6.7	6.2	2003	0.060	1	1.00	1	1.00	1	0.97
Iran	NAME	UM	0.7176	-60.6	-59.0	1990	0.065	2	1.00	2	0.98	2	0.99
Iraq	NAME	UM	0.6539	-23.9	-21.1	1998	0.068	0	0.44	0	0.20	2	0.74
Jordan	NAME	LM	0.7204	-50.6	-46.8	1992	0.052	2	0.99	2	0.89	2	1.00
Kuwait	NAME	H	0.8771	-54.2	-54.7	1990	0.026	2	1.00	2	0.98	2	1.00
Lebanon	NAME	LM	0.7237	-21.1	-19.8	2020	0.091	1	0.98	1	1.00	1	1.00
Libya	NAME	UM	0.7705	-23.8	-15.2	1990	0.077	1	0.93	1	1.00	1	0.90
Morocco	NAME	LM	0.6124	-33.9	-30.6	1990	0.063	0	0.53	0	0.29	0	0.63
Oman	NAME	H	0.7812	-54.1	-47.7	1990	0.067	2	0.80	0	0.31	2	0.85
Palestine	NAME	LM	0.6380	-46.7	-43.4	1990	0.044	2	0.95	2	0.71	2	1.00
Qatar	NAME	H	0.8596	-64.5	-64.6	1992	0.069	2	1.00	2	1.00	2	1.00
Saudi Arabia	NAME	H	0.8284	-33.6	-29.6	1990	0.064	1	0.99	1	1.00	1	1.00
Sudan	NAME	L	0.5731	-43.5	-38.4	1990	0.084	2	0.78	0	0.38	2	0.81
Syrian Arab Republic	NAME	L	0.5887	-43.2	-29.8	1992	0.043	2	0.96	0	0.34	2	0.97
Tunisia	NAME	LM	0.6837	-39.9	-46.1	1990	0.088	2	0.77	2	0.87	2	0.83
Türkiye	NAME	UM	0.7310	-54.7	-50.4	1990	0.057	2	1.00	2	0.93	2	0.88
United Arab Emirates	NAME	H	0.8622	-51.5	-39.1	1990	0.180	2	0.91	0	0.18	2	0.79
Yemen	NAME	L	0.4464	-42.9	-39.0	1990	0.071	2	0.77	0	0.38	2	0.78
Bangladesh	SA	LM	0.5171	22.0	37.2	2012	0.067	1	1.00	1	1.00	1	1.00
Bhutan	SA	LM	0.5243	-21.5	-27.5	1997	0.119	0	0.38	0	0.28	0	0.63
India	SA	LM	0.6070	-19.6	-12.1	1990	0.062	1	1.00	1	1.00	1	1.00
Nepal	SA	LM	0.4710	-21.0	-21.3	1999	0.097	0	0.38	0	0.24	0	0.61
Pakistan	SA	LM	0.5127	-2.1	-2.6	2005	0.113	1	1.00	1	1.00	1	0.99
American Samoa	SEAO	H	0.7430	-9.4	-13.0	1999	0.068	1	0.99	1	0.98	1	0.96
Cambodia	SEAO	LM	0.5000	-26.0	-18.7	1990	0.115	0	0.25	1	0.95	1	0.79
China	SEAO	UM	0.7275	-59.7	-58.1	1990	0.048	2	1.00	2	1.00	2	1.00
Korea, DPR	SEAO	L	0.5771	-5.7	-7.6	2004	0.074	1	1.00	1	1.00	1	0.96
Fiji	SEAO	UM	0.6864	-4.2	-0.3	1998	0.049	1	1.00	1	1.00	1	0.98
Guam	SEAO	H	0.8119	0.0	-15.1	1999	0.027	1	1.00	1	1.00	1	1.00
Indonesia	SEAO	UM	0.6657	-8.5	-15.1	2013	0.069	1	1.00	1	1.00	1	1.00
Kiribati	SEAO	LM	0.5427	-4.5	-5.7	1998	0.080	1	1.00	1	1.00	1	0.99
Lao People's Democratic Republic	SEAO	LM	0.5150	-2.4	-4.6	1994	0.110	1	1.00	1	1.00	1	0.98
Malaysia	SEAO	UM	0.7704	-35.8	-41.1	1990	0.046	0	0.63	2	0.80	2	0.72
Maldives	SEAO	UM	0.6610	-43.4	-26.5	1991	0.060	2	0.97	0	0.36	2	0.87
Marshall Islands	SEAO	UM	0.6055	3.3	4.5	2021	0.102	1	1.00	1	1.00	1	1.00
Mauritius	SEAO	UM	0.7363	-38.1	-37.7	1990	0.022	2	0.90	2	0.93	2	0.74
Micronesia	SEAO	LM	0.5968	-32.8	-32.5	1990	0.078	0	0.41	0	0.26	0	0.51
Myanmar	SEAO	LM	0.5320	-17.0	-17.5	2008	0.106	1	0.93	1	0.98	1	0.88
Nauru	SEAO	H	0.6264	4.8	3.6	2005	0.122	1	1.00	1	1.00	1	0.93
Northern Mariana Islands	SEAO	H	0.7911	-12.4	-13.1	1999	0.068	1	1.00	1	1.00	1	0.99
Palau	SEAO	H	0.7559	-20.6	-21.0	1990	0.068	1	0.88	1	0.92	1	0.80
Papua New Guinea	SEAO	LM	0.4290	-6.6	-8.7	2011	0.103	1	0.99	1	1.00	1	0.99
Philippines	SEAO	LM	0.6694	-5.2	-10.3	2021	0.039	1	1.00	1	1.00	1	1.00
Samoa	SEAO	LM	0.6045	7.6	10.0	2023	0.081	1	1.00	1	1.00	1	1.00
Seychelles	SEAO	H	0.7530	-43.0	-43.0	1991	0.049	2	0.87	2	0.73	2	0.98
Solomon Islands	SEAO	LM	0.4580	15.7	10.7	2023	0.088	1	1.00	1	1.00	1	1.00
Sri Lanka	SEAO	LM	0.7307	-31.7	-30.9	1990	0.040	0	0.69	0	0.34	2	0.89
Taiwan	SEAO	H	0.8815	-58.4	-64.2	1990	0.029	2	1.00	2	1.00	2	0.84
Thailand	SEAO	UM	0.6872	-42.2	-54.2	1995	0.060	2	0.98	2	1.00	2	1.00
Timor-Leste	SEAO	LM	0.4846	-16.7	-23.1	1990	0.108	1	0.97	1	0.96	1	1.00
Tonga	SEAO	UM	0.6360	3.0	3.3	2009	0.076	1	1.00	1	1.00	1	1.00
Tuvalu	SEAO	UM	0.6112	0.3	-4.2	2021	0.082	1	1.00	1	1.00	1	1.00
Vanuatu	SEAO	LM	0.4765	-1.2	-2.8	1994	0.086	1	1.00	1	1.00	1	1.00
Viet Nam	SEAO	LM	0.6430	-31.4	-35.5	1990	0.064	0	0.25	0	0.26	1	0.82
Angola	SSA	LM	0.4991	-16.7	-17.2	1990	0.121	1	0.98	1	0.98	1	0.96

Table S2 continued

Country	Region	Inc.	SDI	Δ DALY	Δ deaths	Peak	w_i	a_i	p_i	a_i^d	p_i^d	a_i^w	p_i^w
Benin	SSA	LM	0.4064	-31.2	-30.9	1990	0.104	1	0.86	0	0.09	1	0.96
Botswana	SSA	UM	0.6776	-26.3	-25.8	2001	0.099	0	0.52	0	0.32	2	0.82
Burkina Faso	SSA	L	0.3067	-16.3	-17.5	1990	0.096	1	0.94	1	0.97	1	0.92
Burundi	SSA	L	0.3072	7.4	12.3	2021	0.141	1	1.00	1	1.00	1	1.00
Cabo Verde	SSA	LM	0.5624	-12.7	-16.0	1995	0.073	1	0.98	1	0.91	1	0.97
Cameroon	SSA	LM	0.4870	-11.7	-5.5	1990	0.098	1	1.00	1	1.00	1	1.00
Central African Republic	SSA	L	0.2591	-3.1	-2.6	1994	0.167	1	0.94	1	0.99	1	0.96
Chad	SSA	L	0.2393	-4.9	0.6	1990	0.115	1	1.00	1	1.00	1	1.00
Comoros	SSA	LM	0.4786	-20.6	-22.2	1991	0.112	1	0.90	1	0.91	1	0.91
Congo	SSA	LM	0.5777	7.0	13.8	2023	0.109	1	1.00	1	1.00	1	1.00
Côte d'Ivoire	SSA	LM	0.4515	-5.5	-6.8	1990	0.106	1	0.99	1	1.00	1	1.00
Democratic Republic of the Congo	SSA	L	0.3950	-0.4	2.4	1990	0.149	1	1.00	1	1.00	1	0.99
Djibouti	SSA	LM	0.4910	-17.3	-18.0	1993	0.136	1	0.98	1	0.97	1	0.98
Equatorial Guinea	SSA	UM	0.6460	-22.2	-23.1	1990	0.140	0	0.58	0	0.42	0	0.58
Eritrea	SSA	L	0.4145	-2.6	-5.5	1990	0.171	1	1.00	1	1.00	1	0.99
Eswatini	SSA	LM	0.6069	-46.3	-39.2	1990	0.128	2	0.88	0	0.52	2	0.90
Ethiopia	SSA	L	0.3937	24.2	29.6	2023	0.122	1	1.00	1	1.00	1	1.00
Gabon	SSA	UM	0.6309	-9.4	-9.0	1991	0.120	1	0.99	1	1.00	1	0.99
Gambia	SSA	L	0.4326	9.1	19.7	2023	0.104	1	1.00	1	1.00	1	1.00
Ghana	SSA	LM	0.5737	-26.8	-26.8	1990	0.084	1	0.82	1	0.91	1	1.00
Guinea	SSA	LM	0.3551	-13.1	-10.0	1990	0.108	1	0.96	1	1.00	1	1.00
Guinea-Bissau	SSA	L	0.3733	13.1	19.6	2023	0.119	1	1.00	1	1.00	1	1.00
Kenya	SSA	LM	0.5486	-13.0	-16.3	2000	0.095	1	0.99	1	0.96	1	0.92
Lesotho	SSA	LM	0.5098	-13.0	-11.1	2002	0.105	0	0.26	0	0.16	2	0.70
Liberia	SSA	L	0.3829	-11.3	-7.4	1990	0.101	1	1.00	1	1.00	1	1.00
Madagascar	SSA	L	0.3799	-14.0	-9.8	1992	0.096	1	0.98	1	1.00	1	1.00
Malawi	SSA	L	0.4094	-15.8	-13.9	2021	0.108	1	0.92	1	0.95	1	0.82
Mali	SSA	L	0.2849	-8.6	-3.1	1990	0.112	1	0.98	1	1.00	1	0.99
Mauritania	SSA	LM	0.5179	-9.2	-5.5	1993	0.103	1	0.99	1	1.00	1	0.92
Mozambique	SSA	L	0.3417	6.0	15.2	2021	0.087	1	1.00	1	1.00	1	1.00
Namibia	SSA	UM	0.6195	-38.5	-36.1	1990	0.101	0	0.55	0	0.34	0	0.43
Niger	SSA	L	0.2030	-4.4	-0.6	2002	0.144	1	0.96	1	1.00	1	0.90
Nigeria	SSA	LM	0.5119	-11.7	-11.2	1991	0.111	1	1.00	1	1.00	1	0.98
Rwanda	SSA	L	0.4621	-25.5	-17.1	1994	0.115	0	0.54	1	0.91	1	0.84
Sao Tome and Principe	SSA	LM	0.5226	-0.2	-3.1	2007	0.093	1	1.00	1	1.00	1	1.00
Senegal	SSA	LM	0.4127	-12.4	-3.7	2001	0.102	1	0.99	1	1.00	1	0.97
Sierra Leone	SSA	L	0.3832	-3.0	-3.0	1990	0.107	1	1.00	1	1.00	1	1.00
Somalia	SSA	L	0.2290	14.1	13.7	2021	0.163	1	1.00	1	1.00	1	0.99
South Africa	SSA	UM	0.6965	-14.4	-14.3	1996	0.049	1	0.98	1	1.00	1	0.90
South Sudan	SSA	L	0.3790	3.7	5.2	2023	0.122	1	1.00	1	1.00	1	1.00
Togo	SSA	L	0.4604	-19.0	-17.5	1990	0.106	1	0.92	1	0.96	1	0.97
Uganda	SSA	L	0.4421	-9.1	-13.9	1994	0.106	1	0.98	1	0.97	1	0.97
United Republic of Tanzania	SSA	LM	0.4658	-3.6	-0.8	2012	0.073	1	1.00	1	1.00	1	1.00
Zambia	SSA	LM	0.5054	8.9	9.1	2011	0.104	1	1.00	1	1.00	1	1.00
Zimbabwe	SSA	LM	0.4911	8.9	-2.3	2003	0.097	1	1.00	1	0.97	1	0.89

S3 Unassigned countries

Table S3 lists the 31 countries that fall below the assignment threshold in the primary analysis, ordered by membership probability, together with their reference cluster under the two-cluster partition. Their median change over the period is -31% , between the two clusters, and their SDI concentrates between 0.55 and 0.70, the band in which the multinomial model of the main text predicts the highest probability of being unassigned.

S4 Sensitivity analyses

S4.1 Spatial weights

Moran's I for the stalled-cluster indicator (and, with two clusters, for its complement) was recomputed with the number of nearest neighbours κ varied from 4 to 10 (Table S4). The statistic moves between 0.53 and 0.58 and no permutation of 999 reached the observed value at any κ .

Table S3: Countries unassigned in the primary analysis. The reference cluster is the label the country carries in the two-cluster partition before the threshold is applied.

Country	Region	SDI	Change (%)	Reference cluster	p_i
El Salvador	Latin America and Caribbean	0.6082	-28.3	2	0.18
Venezuela	Latin America and Caribbean	0.5802	-12.6	2	0.18
Guatemala	Latin America and Caribbean	0.5538	-31.9	2	0.20
Cambodia	Southeast Asia, East Asia and Oceania	0.5000	-26.0	2	0.25
Viet Nam	Southeast Asia, East Asia and Oceania	0.6430	-31.4	2	0.25
Lesotho	Sub-Saharan Africa	0.5098	-13.0	2	0.26
Afghanistan	North Africa and Middle East	0.3337	-29.0	2	0.27
Bolivia	Latin America and Caribbean	0.6317	-36.1	2	0.34
Panama	Latin America and Caribbean	0.7154	-31.4	2	0.35
Nepal	South Asia	0.4710	-21.0	2	0.38
Cuba	Latin America and Caribbean	0.6949	-36.2	2	0.38
Bhutan	South Asia	0.5243	-21.5	2	0.38
Suriname	Latin America and Caribbean	0.6549	-30.1	2	0.40
Saint Vincent and the Grenadines	Latin America and Caribbean	0.6585	-27.8	2	0.40
Micronesia	Southeast Asia, East Asia and Oceania	0.5968	-32.8	2	0.41
Turkmenistan	Central Europe, Eastern Europe and Central Asia	0.6822	-14.0	2	0.43
Iraq	North Africa and Middle East	0.6539	-23.9	2	0.44
Botswana	Sub-Saharan Africa	0.6776	-26.3	2	0.52
Morocco	North Africa and Middle East	0.6124	-33.9	2	0.53
Rwanda	Sub-Saharan Africa	0.4621	-25.5	2	0.54
Namibia	Sub-Saharan Africa	0.6195	-38.5	2	0.55
Algeria	North Africa and Middle East	0.6578	-39.2	2	0.57
Equatorial Guinea	Sub-Saharan Africa	0.6460	-22.2	2	0.58
Azerbaijan	Central Europe, Eastern Europe and Central Asia	0.7053	-32.5	2	0.60
Malaysia	Southeast Asia, East Asia and Oceania	0.7704	-35.8	2	0.63
Belize	Latin America and Caribbean	0.6285	-32.2	2	0.64
Bosnia and Herzegovina	Central Europe, Eastern Europe and Central Asia	0.7129	-40.7	2	0.64
Albania	Central Europe, Eastern Europe and Central Asia	0.7068	-41.5	2	0.65
Bulgaria	Central Europe, Eastern Europe and Central Asia	0.7708	-34.8	2	0.66
Sri Lanka	Southeast Asia, East Asia and Oceania	0.7307	-31.7	2	0.69
Belarus	Central Europe, Eastern Europe and Central Asia	0.8036	-23.1	2	0.69

Table S4: Moran’s I of assigned membership by number of nearest neighbours. Six neighbours is the primary choice.

Neighbours κ	Moran’s I	Permutation p
4	0.584	0
5	0.574	0
6	0.558	0
8	0.554	0
10	0.534	0

S4.2 Tuning constants

The framework has two constants: π , the minimum median membership probability every cluster must reach for a partition to be accepted (main-text equation (10)), and τ , the membership probability a country must reach to be reported as assigned (equation (9)). Table S5 shows the consequence of varying each over a grid, holding the other at its primary value. The number of clusters is 2 for every π from 0.75 to 0.90. At $\pi = 0.70$ the rule admits $k = 3$, because the weakest cluster of the three-cluster partition has a median membership of exactly 0.70; this coincidence is the reason the rule is set above the assignment threshold rather than equal to it. Varying τ moves countries between the assigned and unassigned classes but never changes the label of an assigned country, since labels come from the reference partition.

S4.3 Number of clusters under every robustness run

Table S6 gives the three selection diagnostics for every k from 2 to 6 in each of the five analyses summarised in main-text Table 5. The pattern is the same in the primary, deaths, window and bootstrap runs: silhouette falls from $k = 2$, the gap statistic peaks at 3 or 4, and the weakest cluster’s median membership drops below 0.80 at $k = 3$. The narrow-interval subset is the exception: the gap

Table S5: Sensitivity to the tuning constants. Left, number of clusters selected as π varies with $\tau = 0.70$. Right, assignment counts as τ varies with $k = 2$.

π	Selected k	τ	Unassigned	Assigned stalled	Assigned decline
0.7	3	0.6	23	78	100
0.75	2	0.7	31	78	92
0.8	2	0.8	39	78	84
0.85	2	0.9	55	73	73
0.9	2				

statistic finds no structure, while stability supports three clusters with medians above 0.85.

Table S6: Selection diagnostics for every analysis. Gap statistic with standard error from 100 reference sets; membership probabilities from 1,000 draws.

Analysis	k	Silhouette	Gap	Weakest median p	Share unassigned
DALYs 1990 to 2023 (primary)	2	0.442	0.530 (0.024)	0.96	0.15
	3	0.389	0.608 (0.016)	0.70	0.18
	4	0.298	0.617 (0.015)	0.53	0.31
	5	0.306	0.588 (0.015)	0.01	0.32
	6	0.279	0.580 (0.015)	0.03	0.29
Deaths 1990 to 2023	2	0.443	0.529 (0.029)	0.93	0.18
	3	0.380	0.609 (0.021)	0.66	0.28
	4	0.338	0.605 (0.017)	0.71	0.26
	5	0.238	0.604 (0.016)	0.56	0.43
	6	0.225	0.610 (0.016)	0.64	0.40
DALYs 2000 to 2023	2	0.487	0.561 (0.030)	0.99	0.06
	3	0.432	0.563 (0.020)	0.44	0.16
	4	0.327	0.589 (0.015)	0.63	0.31
	5	0.259	0.627 (0.013)	0.70	0.38
	6	0.242	0.623 (0.013)	0.31	0.35
DALYs, narrowest half (n = 101)	2	0.343	0.388 (0.034)	0.86	0.26
	3	0.299	0.394 (0.030)	0.85	0.29
	4	0.274	0.393 (0.025)	0.52	0.29
	5	0.288	0.411 (0.025)	0.26	0.24
	6	0.279	0.583 (0.016)	0.31	0.21
Country bootstrap	2	0.442	0.535 (0.025)	0.96	0.12
	3	0.389	0.613 (0.020)	0.73	0.11
	4	0.298	0.620 (0.018)	0.75	0.18
	5	0.306	0.590 (0.016)	0.19	0.18
	6	0.279	0.583 (0.016)	0.31	0.21

S4.4 The narrow-interval solution

Table S7 gives the composition by super region of the three-cluster solution on the 101 countries with mean interval width below the median, and the membership of each cluster. Cluster 1 (stalled) is Latin American and Pacific; cluster 2 is the post-Soviet and Eastern European rise and fall; cluster 3 is the high-income steady decline. The pooled analysis of all 201 countries absorbs clusters 2 and 3 into one sustained-decline cluster.

Membership of the three clusters:

Cluster 1 (stalled) Bahamas; Guam; Panama; Nicaragua; Philippines; Uzbekistan; Dominica; Dominican Republic; El Salvador; Fiji; Guatemala; Jamaica; Mexico; Paraguay; South Africa; Venezuela.

Cluster 2 (rise and fall) Bulgaria; Croatia; Greece; Guyana; Kuwait; Latvia; Lithuania; Romania; Russian Federation; Seychelles; Slovakia; Kyrgyzstan; Palestine; Sri Lanka; Syrian Arab Republic; Albania; Armenia; Azerbaijan; Belarus; Belize; Bosnia and Herzegovina;

Table S7: Composition of the three-cluster solution on the narrowest half of intervals.

Super region	Cluster 1 (stalled)	Cluster 2 (rise and fall)	Cluster 3 (steady decline)	Unassigned	Total
Central Europe, Eastern Europe and Central Asia	1	22	4	1	28
High income	0	1	20	9	30
Latin America and Caribbean	11	2	0	16	29
North Africa and Middle East	0	3	1	0	4
Southeast Asia, East Asia and Oceania	3	2	1	3	9
Sub-Saharan Africa	1	0	0	0	1
Total	16	30	26	29	101

Georgia; Kazakhstan; Mongolia; Montenegro; North Macedonia; Republic of Moldova; Serbia; Turkmenistan; Ukraine.

Cluster 3 (steady decline) Australia; Austria; Bahrain; Belgium; Czechia; Denmark; Estonia; Finland; Iceland; Ireland; Israel; Luxembourg; Malta; Netherlands; New Zealand; Norway; Poland; Portugal; Republic of Korea; Singapore; Slovenia; Spain; Sweden; Switzerland; United Kingdom; China.

S5 Additional figures

Figure S1 plots the selection diagnostics of Table S6 for the four robustness runs, in the same form as main-text Figure 3. In three of the four the picture matches the primary analysis: a steep fall in the weakest cluster median membership between $k = 2$ and $k = 3$ while silhouette declines smoothly. The narrow-interval subset is the exception, with medians of 0.86 and 0.85 at $k = 2$ and $k = 3$ before the fall, which is the three-cluster solution described in Section S4.4.

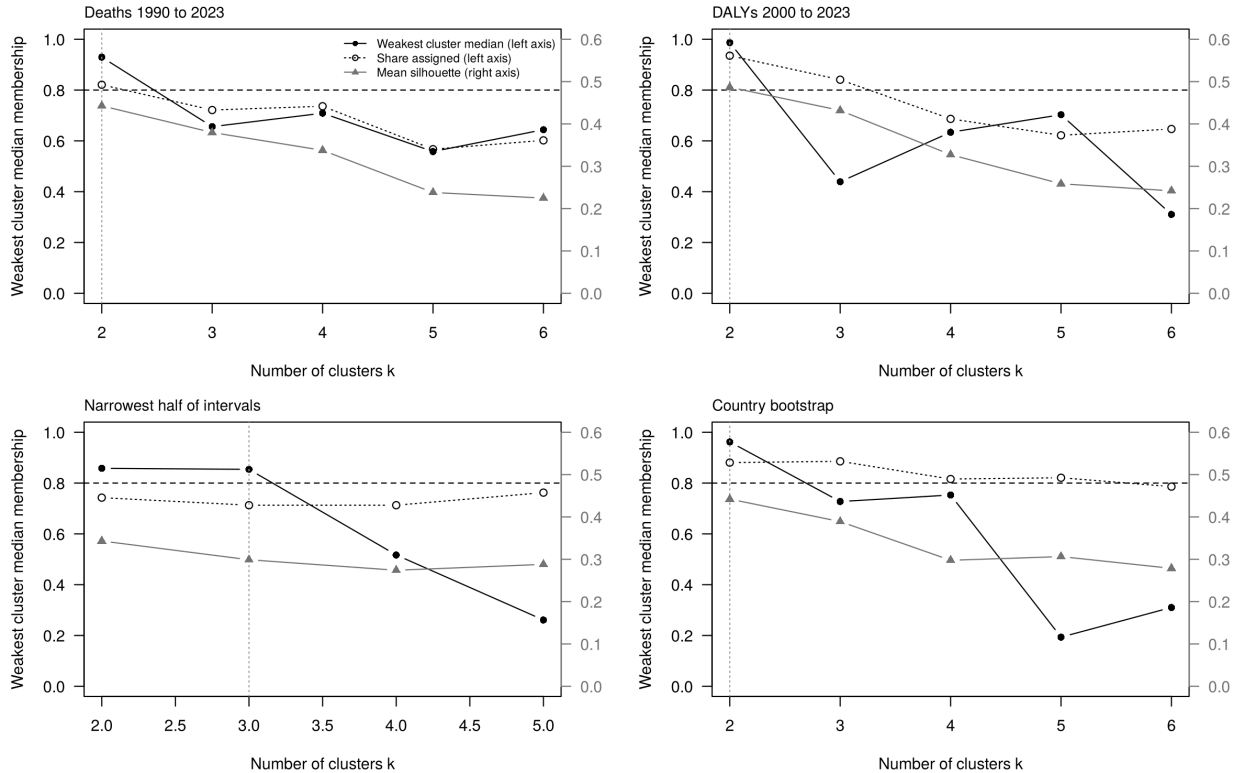


Figure S1: Selection diagnostics for the four robustness runs. Filled circles, weakest cluster median membership probability (left axis); open circles, share of countries assigned (left axis); triangles, mean silhouette width (right axis). Dashed line, the 0.80 rule; dotted vertical, the k the rule selects.

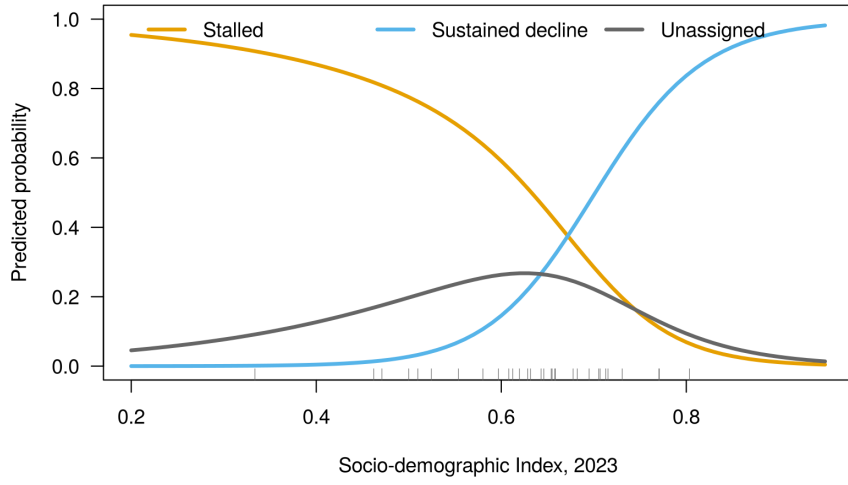


Figure S2: Predicted probability of each membership outcome across the Socio-demographic Index from the multinomial model on SDI alone. Ticks on the horizontal axis mark the SDI of the 31 unassigned countries. Referred to in main-text Section 4.3.

S6 Reproduction

The analysis runs in R 4.3.3 with the packages `cluster`, `nnet` and `maps`, all available from CRAN or from Ubuntu's `r-cran-*` repositories; no other packages are needed. The supplementary archive contains:

`data/panel_cvd.csv` age-standardised and all-ages cardiovascular DALY rates with intervals, 203 countries by 34 years.

`data/panel_cvd_deaths.csv` the same for death rates.

`data/sdi_2023.csv`, `superregion.csv`, `income_group.csv` covariate lookups.

`data/cvd_case_study_membership.csv` the country list and metadata that fixes the analysis set.

`analysis/final_analysis.R` one run from one seed (20261003) with $B = 1,000$ that produces every table and figure in the main text and this supplement; about 75 seconds on a laptop.

`analysis/build_supplement_tables.R` renders the supplementary tables from the saved workspace.

`analysis/unwrap.R`, `unwrap_csv.R` helpers that convert connector responses into the panel files, for anyone repeating the extraction.

The core of the framework, main-text equations (6) to (8), is the following function, reproduced from `final_analysis.R`:

```
stability <- function(Y, S, hc, k, B, mode = "value") {
  ref <- cutree(hc, k); shape <- sweep(Y, 1, rowMeans(Y))
  agree <- numeric(nrow(Y)); drawn <- numeric(nrow(Y))
  for (b in 1:B) {
    if (mode == "value") {
      Yb <- Y + matrix(rnorm(length(Y)), nrow(Y)) * S # equation (6)
      idx <- 1:nrow(Y)
      clb <- cutree(hclust(dist(sweep(Yb, 1, rowMeans(Yb))), "ward.D2"), k)
    } else { # country bootstrap
      idx <- sort(unique(sample(nrow(Y), replace = TRUE)))
      clb <- cutree(hclust(dist(shape[idx, ]), "ward.D2"), k)
    }
    lab <- apply(table(clb, ref[idx]), 1, which.max) # equation (7)
    agree[idx] <- agree[idx] + (lab[clb] == ref[idx])
    drawn[idx] <- drawn[idx] + 1
  }
  list(ref = ref, p = agree / drawn) # equation (8)
}
```

The uncertainty aware choice of k (equation (10)) is then one line: the largest k for which `min(tapply(p, ref, median)) >= 0.80`.